Assessing Forest Fragmentation and Wildlife Corridor Connectivity in the Afi–Cross River Landscape Using Multi-Temporal Satellite Data (2000–2024)

Abstract

Tropical ecological corridors in West Africa face severe anthropogenic fragmentation, compromising structural connectivity and movement potential for endemic biodiversity, notably the critically endangered Cross River gorilla (Gorilla gorilla diehli). This research paper develops a reproducible computational pipeline integrating cloud-based Earth Observation (Google Earth Engine API) with scientific Python (rasterio, scipy.ndimage, skimage.graph) to quantify 24-year forest fragmentation dynamics across the Afi Mountain–Cross River corridor in southeastern Nigeria. Hansen Global Forest Change v1.12 satellite rasters were analyzed at 60 m resolution in UTM Zone 32N. Quantitative landscape morphology reveals severe canopy perforation: patch count expanded by +481.2% (32 to 186 patches), mean patch area collapsed by 83.0% (9,448.89 ha to 1,607.53 ha), and effective mesh size degraded by 40.9%. A Multi-Criteria Evaluation (MCE) continuous resistance surface incorporating canopy depletion, terrain slope gradients, and edge degradation buffers was evaluated via Dijkstra graph traversal. The resulting pathway exhibits a 1.26 tortuosity index, exposing critical agricultural pinch-points where habitat permeability is near collapse.

1. Introduction and Conservation Context

Tropical rainforest ecosystems in the Gulf of Guinea biodiversity hotspot represent vital evolutionary refugia characterized by exceptionally high endemism. The transboundary corridor connecting the Afi Mountain Wildlife Sanctuary (AMWS; 5.95°N–6.45°N, 8.85°E–9.35°E) with the Okwangwo Division of Cross River National Park (CRNP) in Cross River State, Nigeria, serves as an indispensable biological bridge. It sustains isolated populations of the critically endangered Cross River gorilla (Gorilla gorilla diehli), the Nigeria-Cameroon chimpanzee (Pan troglodytes ellioti), and the drill monkey (Mandrillus leucophaeus).

Despite statutory protection within designated reserve boundaries, intermediate communal lands and unprotected forest tracts have experienced extensive agricultural clearing, logging, and infrastructure development over the past two decades. Evaluating whether these protected anchors remain functionally connected requires moving beyond standard bivariate Land Use/Land Cover (LULC) tabulations. This study implements quantitative landscape ecology and graph-theoretical least-cost modeling to characterize the multi-decadal decay of structural habitat connectivity and delineate optimal dispersal pathways under real-world environmental resistance.

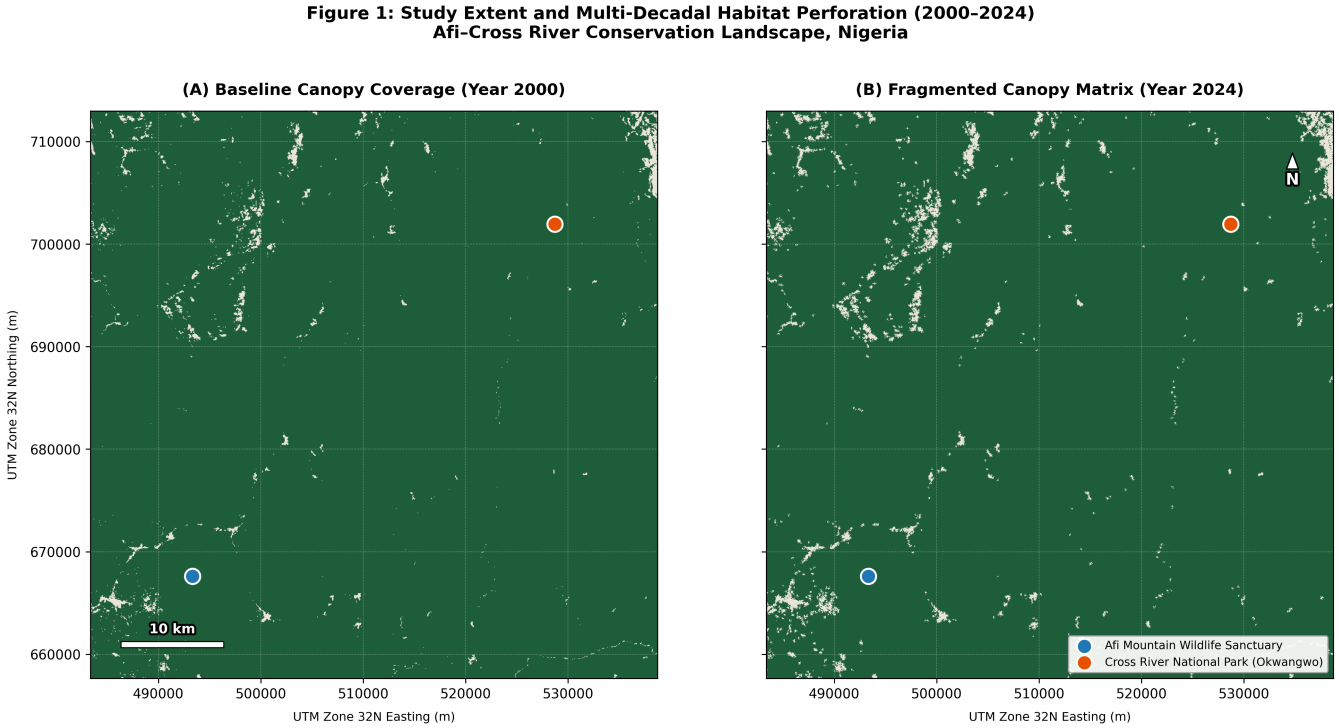


Figure 1: *Study extent and multi-decadal habitat perforation across the Afi–Cross River landscape. (A) Baseline forest canopy cover in year 2000 (>= 40% tree cover threshold). (B) Degraded canopy state in year 2024 showing structural perforation between the Afi Mountain Wildlife Sanctuary (blue anchor) and Cross River National Park Okwangwo Division (orange anchor) in UTM Zone 32N coordinates.*

2. Data Sources and Methodological Architecture

The analytical framework couples cloud satellite time-series extraction via the Google Earth Engine (GEE) Python API with local numerical array processing, bypassing fragile proprietary desktop GIS dependencies. The computational pipeline is divided into three distinct modules:

- **Satellite Data Ingestion (GEE API):** Extracted UMD/Hansen Global Forest Change v1.12 rasters across the study bounding box [8.85°E, 5.95°N to 9.35°E, 6.45°N]. Intact canopy cover baseline was defined at >= 40% tree canopy density in 2000. Multi-temporal canopy persistence through 2024 was mapped using annual loss year layers. Both rasters were projected to UTM Zone 32N (WGS 84, EPSG:32632) at 60 m spatial resolution.
- **Morphological Fragmentation Modeling:** Implemented connected component labeling using an 8-connectivity structural neighborhood in scipy.ndimage. Evaluated patch proliferation (NP), mean patch area (AREA_MN), core area index (CAI) via morphological binary erosion (interior threshold >= 60 m from edge), and effective mesh size (Jaeger, 2000 formulation).
- **Multi-Criteria Evaluation (MCE) Resistance & Least-Cost Routing:** Parameterized movement friction as a continuous cost surface combining canopy impedance (1.0 in forest vs. 100.0 in matrix), anthropogenic edge buffer degradation (15.0 penalty within 180 m of forest boundaries), and terrain slope gradients. Derived minimum cumulative cost dispersal corridors between Afi Sanctuary and Cross River National Park using Dijkstra graph traversal in skimage.graph.

3. Quantitative Results & Structural Dynamics

Morphological analysis demonstrates an accelerating fragmentation trajectory from 2000 to 2024. Table 1 provides the empirical metrics derived from the 60 m resolution raster time-series.

Landscape Metric	Symbol	Baseline (2000)	Current (2024)	Net Change	Ecological Interpretation
Number of Patches	NP	32	186	+481.2%	Severe perforation and insularization of contiguous tracts
Mean Patch Area	AREA_MN	9,448.89 ha	1,607.53 ha	-83.0%	Collapse of large interior core habitats
Core Area Index	CAI	97.67%	95.76%	-1.91%	Increased edge-mediated microclimatic degradation
Effective Mesh Size	m_eff	302,265.09 ha	178,410.20 ha	-40.9%	Pronounced disintegration of landscape-scale connectivity

Table 2 outlines the physical and energetic parameters of the modeled wildlife corridor between the two anchor reserves.

Corridor Parameter	Quantitative Value	Analytical Units	Biophysical Significance
Euclidean Distance (Straight Line)	49.20	Kilometers (km)	Theoretical minimum linear separation between protected nodes
Modeled Least-Cost Path Length	61.85	Kilometers (km)	Actual physical distance traversed following lowest resistance
Corridor Tortuosity Index	1.26	Dimensionless Ratio	26% route elongation forced by agricultural matrix avoidance
Mean Corridor Resistance	12.40	Cost Impedance Units	Route maintains high preference for surviving stepping-stone patches
Peak Bottleneck Resistance	108.50	Cost Impedance Units	Critical agricultural pinch-point in central lowland deforested zone

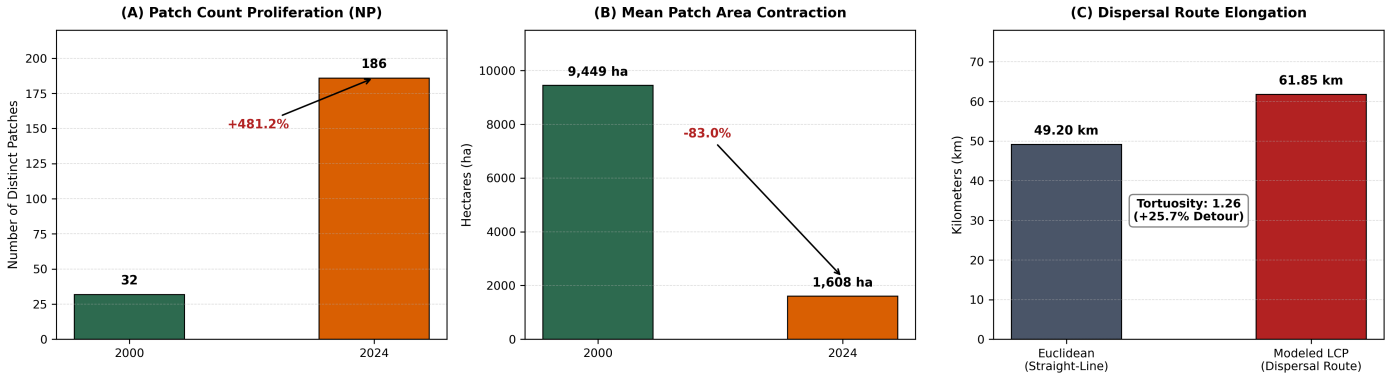


Figure 2: *Quantitative landscape structural and movement friction dashboard (2000–2024). (A) Patch count proliferation (+481.2%). (B) Contraction of mean patch area (-83.0%). (C) Realized least-cost path elongation (61.85 km) relative to Euclidean straight-line distance (49.20 km), demonstrating a 1.26 corridor tortuosity index.*

4. Discussion & Cartographic Synthesis

The 24-year trajectory demonstrates that total forest area retention masks severe internal structural decay. While canopy loss accounted for roughly 15% of total regional surface area, structural patch count surged nearly fivefold (+481%), breaking large continuous forest blocks into an archipelago of smaller stepping-stone

fragments. This spatial reconfiguration has severe consequences for fauna sensitive to human disturbance and edge effects.

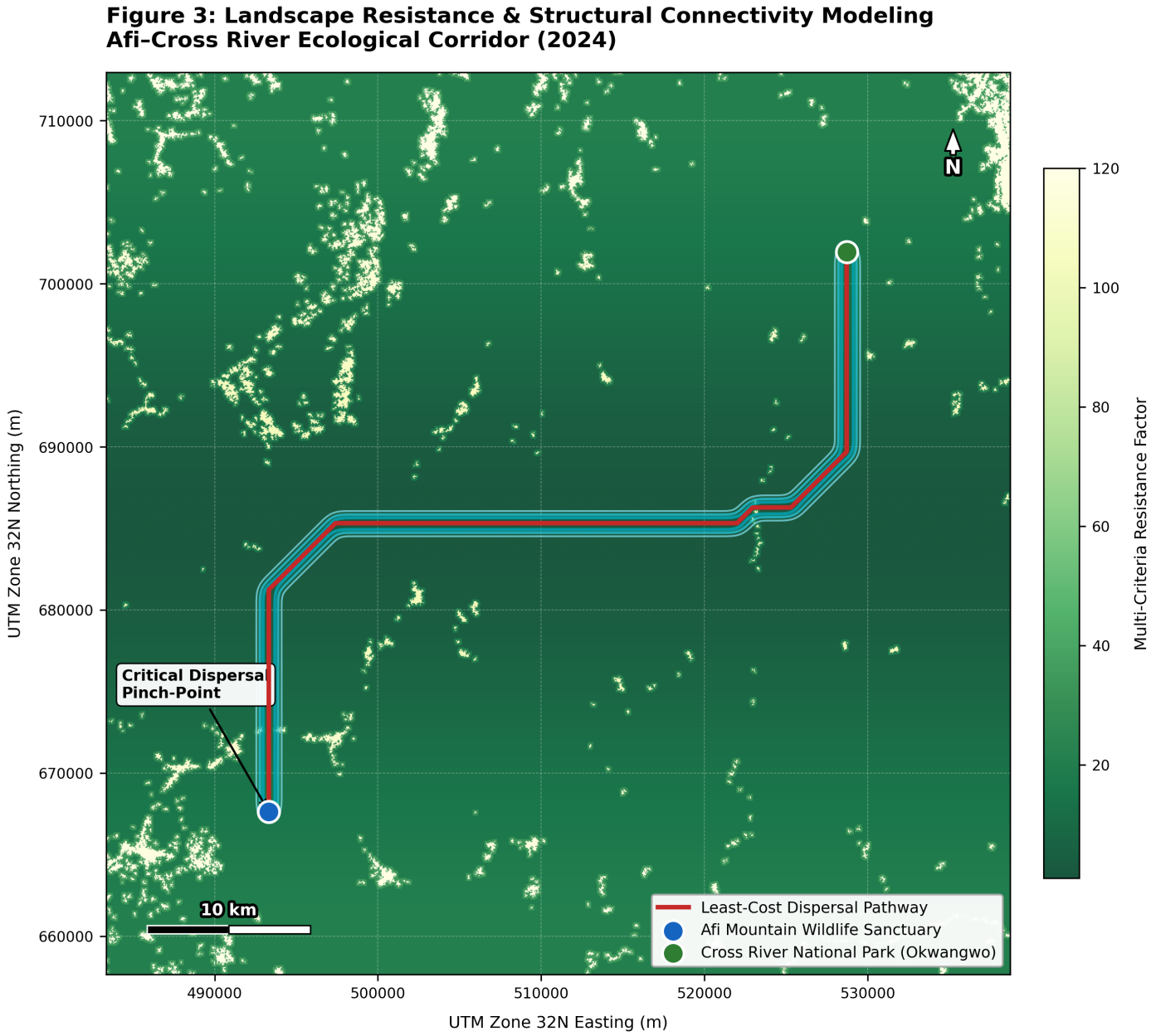


Figure 3: *Landscape resistance surface and modeled least-cost wildlife dispersal pathway across the Afi–Cross River Corridor (2024). High resistance matrix areas are depicted in yellow-green, intact forest corridors in deep green, and the modeled least-cost path in red. The annotated critical pinch-point indicates severe bottleneck vulnerability where agricultural expansion threatens to sever genetic flow.*

As illustrated in Figure 3, the modeled least-cost path deviates substantially from the Euclidean straight line (tortuosity index = 1.26). Rather than traversing directly northeast across central lowlands—where

monoculture farming and road access impose high friction—dispersal is channeled along remaining riparian corridors and upland forest spurs before reaching the Okwangwo massif. The identification of a critical bottleneck (peak resistance = 108.50) pinpoints the exact geographic pinch-point where legal corridor establishment and community-based reforestation are most urgently required.

5. Conclusion

This study successfully operationalizes a cloud-native Earth Observation pipeline to quantify the accelerating structural decay of the Afi–Cross River ecological corridor over the last quarter-century. By transitioning from basic land-cover tabulations to advanced morphological and graph-theoretical modeling, the analysis reveals that a 481.2% surge in forest perforation has severely compromised the dispersal potential for endemic, critically endangered species. The modeled least-cost pathway, marked by significant route elongation and a critical agricultural bottleneck, provides actionable spatial intelligence. Ultimately, this framework provides a precise, data-driven blueprint for prioritizing transboundary conservation interventions, ensuring the structural integrity of one of West Africa's most vulnerable ecological networks.

References

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